



More graphs of linear equations

What to do

- What is a linear function?
- Use the gradient and y-intercept to sketch graphs of the following equations.
 - $2x - y = -1$
 - $3x = 5$
 - $3y = 4 - 2x$
 - $4y + x = -11$
- Use x - and y -intercepts to sketch graphs of the following equations.
 - $2y + x = 4$
 - $y = 5 - \frac{1}{2}x$
 - $3y + 2x = 12$
 - $3x - 4y = -12$
 - $\frac{x}{2} + \frac{y}{3} = 1$
 - $\frac{2}{3}x + \frac{1}{3}y = -4$
- Find the value of b given that the point $(1, 5)$ lies on the line $2y = b - \frac{7}{2}x$.
 - What is the x -intercept?
- If a linear graph has an x -intercept of 4 and a y -intercept of -2 , is the gradient positive or negative? Explain.
 - If a linear graph has an x -intercept of -2 and a y -intercept of -3 , is the gradient positive or negative? Explain.
- What is the slope of a horizontal line? Sketch a graph and verify your answer.
- What is the slope of vertical line? Sketch a graph and verify your answer.
- Sketch graphs for each of the following equations.
 - $2y - 1 = -2$
 - $\frac{y}{3} = 0$
 - $\frac{-3}{2}x = -6$
- Graph each of these functions for the given domain. State the range for each function.
 - $f(x) = 2x + 1, -2 \leq x \leq 3$
 - $f(x) = 5 - x, -3 \leq x \leq 2$
 - $f(x) = -4x - 1, -2 < x < 2$
- Given the following graphs, state the range.
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